

Rupali Patel

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Education

VIT Bhopal University

B.Tech Electronics and Communication Engineering

Cumulative GPA: 9.25/10

Bhopal, Madhya Pradesh

Expected June 2027

Technical Skills and Tools

Language: C++, Java

Tools: LTspice, Proteus

Coursework: Embedded System, Iot, Signal Processing

Projects

Pulse Oximeter

July 2024- Sep 2024

- **Description:** Designed and implemented a pulse oximeter system using Arduino with integrated sensor interfacing to measure heart rate and blood oxygen (SpO₂) levels in real time.
- Developed an embedded system to acquire physiological signals from the MAX30100 sensor and process them using Arduino.
- Implemented real-time display of heart rate and SpO₂ values on an LCD module for easy user interaction.
- Ensured reliable I2C communication between Arduino and MAX30100 sensor module.
- Performed testing and validation of sensor readings to improve accuracy and stability of measurements.
- **Technology:** Arduino, MAX30100, LCD, Embedded C, I2C Protocol, Basic Electronic Components.

Object Detection In Water

Sep 2024- Dec 2024

- **Description:** Developed an Arduino-based underwater object detection system to sense and identify submerged objects using connected sensors and control electronics.
- Designed and implemented a real-time detection mechanism to measure distance and presence of objects in water environments.
- **Technology:** Arduino IDE, Ultrasonic Sensor, Embedded C, Java (for monitoring/interface), Serial Communication, Basic Electronic Components.
- **Role :** Hardware Designer – designed the complete circuit, selected components, and handled sensor interfacing with Arduino.

Sign Language to Speech Conversion System

Jan 2025-Apr 2025

- **Description:** Created a system that translates sign language into text and speech, enabling effective communication between deaf/mute individuals and non-sign language users.
- Designed a sensor-based hand gesture input mechanism using flex sensors to capture finger movements.
- Enabled wireless data transmission of recognized gestures using the HC-05 Bluetooth module.
- **Technology:** Arduino IDE, C++, Flex Sensors, HC-05 Bluetooth Module, Serial Communication, Basic Electronic Components.
- **Role:** Hardware integration of flex sensors and Bluetooth module with the microcontroller.
- Testing, debugging, and troubleshooting of hardware and communication issues.

Experience

Embedded System Design Internship-Maven Silicon

Jan 2025-March 2025

- Developed embedded firmware and hardware interfacing for **Alert system for women's** safety project.
- Improved skill level in microcontroller-based system design and embedded programming.
- **Link-** https://drive.google.com/file/d/1m3kuM5QLx20ST_MdDpKuo79MoKr2HIyt/view?usp=sharing

Achievements

- Awarded with Dreamed STAR Scholarship from VIT Bhopal.
- Secured first position in district merit in class 12th.
- Awarded with **Reliance Foundation Scholarship** for academic excellence.

Additional Information

Languages: Hindi, English

Hobbies: Gaming, Reading books, Sketching